

Clique Partitions of Chordal Graphs

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Abstract

We derive the asymptotic clique partition theorem for chordal graphs from the fractional triangle theorem proved in Paper C and the fixed-family fractional-to-integral packing theorem of Haxell–Rödl and Yuster. Specifically, for every chordal graph G on n vertices,

$$\text{cp}(G) \leq \left(\frac{1}{6} + o(1)\right) n^2.$$

The constant is sharp: the complete split graph $K_p \vee \overline{K}_{2p}$, analyzed in Paper A, is chordal and has clique partition number $n^2/6 + O(n)$. Thus

$$\max\{\text{cp}(G) : G \text{ chordal}, |V(G)| = n\} = \left(\frac{1}{6} + o(1)\right) n^2.$$

This preprint contains the short final rounding and sharpness step of the series.

Contents

1	Introduction	1
2	Triangle packings and clique partitions	2
3	Imported fractional theorem	2
4	Fractional-to-integral triangle rounding	3
5	Upper bound	3
6	Sharpness	3

7	Proof-status audit	4
8	Discussion	4
9	Further directions	4

1 Introduction

The edge clique partition number $\text{cp}(G)$ is the minimum number of cliques whose edge sets partition $E(G)$. Erdős, Ordman and Zalcstein asked whether every n -vertex chordal graph admits an edge clique partition with

$$\frac{n^2}{6} + O(n)$$

parts. The complete split graph $K_p \vee \overline{K}_{2p}$, $n = 3p$, shows that the constant $1/6$ is the correct asymptotic barrier.

This preprint is the final step in the series. Paper A proves the sharp split graph theorem. Paper B develops the local and structural chordal toolkit. Paper C proves the fractional chordal theorem

$$|E(G)| - 2\nu_3^*(G) \leq \left(\frac{1}{6} + o(1)\right)n^2.$$

Here we combine that theorem with standard fractional-to-integral packing rounding for fixed graph families.

Main result

Theorem 1.1 (Chordal Clique Partition Theorem). *For chordal graphs,*

$$\max\{\text{cp}(G) : G \text{ chordal}, |V(G)| = n\} = \left(\frac{1}{6} + o(1)\right)n^2.$$

2 Triangle packings and clique partitions

Let $\nu_3(G)$ denote the maximum number of edge-disjoint triangles in G , and let $\nu_3^*(G)$ denote the maximum total weight of a fractional triangle packing. Thus

$$\nu_3^*(G) = \max \sum_{T \in K_3(G)} \psi_T$$

subject to

$$\sum_{T \ni e} \psi_T \leq 1 \quad (e \in E(G)), \quad \psi_T \geq 0.$$

Lemma 2.1 (Triangle packing to clique partition). *For every graph G ,*

$$\text{cp}(G) \leq |E(G)| - 2\nu_3(G).$$

Proof. Start with the partition of $E(G)$ into singleton-edge cliques, which has $|E(G)|$ parts. For each triangle in an edge-disjoint triangle packing, replace its three singleton-edge cliques by one triangle clique. This saves two cliques per packed triangle. Applying this to a maximum triangle packing gives the inequality. $\square$

3 Imported fractional theorem

The main input from Paper C is the following.

Theorem 3.1 (Fractional Chordal Theorem, Paper C). *For every chordal graph G on n vertices,*

$$|E(G)| - 2\nu_3^*(G) \leq \left(\frac{1}{6} + o(1)\right) n^2.$$

Paper C proves this theorem using the clique-tree decomposition and local toolkit developed in Paper B.

4 Fractional-to-integral triangle rounding

We use the standard fixed-family fractional-to-integral packing theorem of Haxell–Rödl and Yuster. In the special case needed here, it says the following.

Theorem 4.1 (Fractional-to-integral triangle rounding). *For every n -vertex graph G ,*

$$\nu_3(G) \geq \nu_3^*(G) - o(n^2).$$

Remark 4.2. The general theorem applies to any fixed finite family of graphs. Here we apply it to the one-element family $\{K_3\}$.

5 Upper bound

Let G be chordal on n vertices. By Lemma 2.1,

$$\text{cp}(G) \leq |E(G)| - 2\nu_3(G).$$

By Theorem 4.1,

$$\nu_3(G) \geq \nu_3^*(G) - o(n^2),$$

so

$$\text{cp}(G) \leq |E(G)| - 2\nu_3^*(G) + o(n^2).$$

Using Theorem 3.1, we get

$$\text{cp}(G) \leq \left(\frac{1}{6} + o(1)\right) n^2.$$

Thus

$$\max\{\text{cp}(G) : G \text{ chordal}, |V(G)| = n\} \leq \left(\frac{1}{6} + o(1)\right) n^2.$$

6 Sharpness

The lower bound is supplied by the split graph construction from Paper A.

Theorem 6.1 (Split lower bound, Paper A). *Let*

$$G_p = K_p \vee \overline{K}_{2p}.$$

Then G_p is split, hence chordal, has $n = 3p$ vertices, and satisfies

$$\text{cp}(G_p) \geq \frac{n^2}{6} - O(n).$$

Since every split graph is chordal, Theorem 6.1 gives

$$\max\{\text{cp}(G) : G \text{ chordal}, |V(G)| = n\} \geq \frac{n^2}{6} - O(n).$$

Combining the lower bound with the upper bound proves Theorem 1.1.

7 Proof-status audit

Item	Source	Status
Split lower bound	Paper A	preprint
Local chordal toolkit	Paper B	preprint; includes local toolkit and certificate usage
Fractional chordal theorem	Paper C	preprint; imports Paper B
Fractional-to-integral rounding	external theorem	standard fixed-family rounding theorem
Chordal clique partition theorem	Paper D	preprint; follows from above

8 Discussion

Paper D is intentionally short. The difficult parts of the argument are not in this paper: the split extremal construction is treated in Paper A, and the fractional chordal theorem is proved in Papers B and C. The role of this preprint is to make explicit that the fractional result is strong enough to imply the integral clique partition theorem asymptotically.

A possible final presentation choice is to merge this preprint into the end of Paper C. Keeping Paper D separate has the advantage of cleanly distinguishing the fractional chordal theorem from the final clique partition corollary.

9 Further directions

The final theorem leaves several natural follow-up questions.

- (i) **Exact finite bounds.** The result is asymptotic. Determining the best possible $O(n)$ term, or exact extremal values for finite n , remains open.
- (ii) **Constructive partitions.** The proof route is through fractional triangle packing and asymptotic rounding. It would be useful to extract explicit near-optimal clique partitions for chordal graphs.
- (iii) **Shorter proofs for subclasses.** The split case has a much cleaner proof, and intermediate chordal subclasses may admit simpler arguments.
- (iv) **Presentation choice.** Paper D may ultimately remain standalone or be merged into Paper C as the final rounding section.

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